

Computational Methods in Macroeconomics

Topic 4: Application

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Application

- ▶ Application of what we have learned so far:
 1. Extend the labour-leisure choice model
 2. Solve and calibrate it
 3. Conduct counterfactual and welfare analyses
- ▶ Coursework: Extend this exercise

Why Do Americans Work So Much More than Europeans?

OUTPUT, LABOR SUPPLY, AND PRODUCTIVITY				
output = hours worked $\times$ productivity				
		<i>Output</i>	<i>Hours Worked</i>	<i>Productivity</i>
1993–96	Germany	74	75	99
	France	74	68	110
	Italy	57	64	90
	Canada	79	88	89
	United Kingdom	67	88	76
	Japan	78	104	74
	United States	100	100	100
1970–74	Germany	75	105	72
	France	77	105	74
	Italy	53	82	65
	Canada	86	94	91
	United Kingdom	68	110	62
	Japan	62	127	49
	United States	100	100	100

Prescott (2004): mainly because European labour income is taxed at a higher rate

Road Map

1. Extend the Labour-Leisure Choice Model (Ch 4.2)
 - ▶ Add a government (also Ch 2.3)
 - ▶ Add a production side (also Ch 2.4)
2. Map the Model to the Data (Ch 4.3-4.4) [\[+\]](#)
3. Calibration (Ch 4.7) [\[+\]](#)
4. Prediction and Welfare Analysis (Ch 4.5, Ch 2.5) [\[+\]](#)

Extend the Labour-Leisure Choice Model

Labour-Leisure Choice Model

- ▶ Our model: a worker chooses labour and leisure
- ▶ Each country is represented by a worker
- ▶ 100 hours of non-sleeping time per week
- ▶ Utility function:

$$\begin{aligned} & U(c) + V(100 - h) \\ &= \log c + \alpha \log(100 - h) \end{aligned}$$

- ▶ Extend the model by adding a government and a production side

Government

- ▶ In each country, the government does the following:
 - ▶ raise revenue to finance government expenditure g
 - ▶ provide lump-sum transfer payments λ
 - ▶ tax labour income at the rate τ_h and consumption at the rate τ_c
- ▶ Government budget constraint (GBC):

$$\underbrace{g + \lambda}_{\text{expenditure}} = \underbrace{\tau_c c + \tau_h wh}_{\text{tax revenue}}$$

Production

- ▶ Output in a country is produced according to a Cobb-Douglas production function:

$$y = zk^{\theta}h^{1-\theta}$$

where

- ▶ z : country-specific level of total factor productivity (TFP)
 - ▶ k : capital stock
 - ▶ h : labour input
- ▶ For simplicity, assume that k is a fixed number in each country

Firm's Problem

- ▶ Firm hires labour h to maximise its profits
- ▶ r : rental rate of capital
- ▶ w : wage rate
- ▶ The firm's profit maximisation problem:

$$\max_h \underbrace{zk^\theta h^{1-\theta}}_{\text{sales}} - \underbrace{(rk + wh)}_{\text{costs}}$$

- ▶ Note that k is fixed

Firm's Problem: First-Order Condition

- The first-order condition (FOC) for labour:

$$\frac{d}{dh} [zk^{\theta}h^{1-\theta} - rk - wh] = 0$$

$$\Rightarrow (1 - \theta)zk^{\theta}h^{-\theta} - w = 0$$

$$\Rightarrow w = \underbrace{(1 - \theta)zk^{\theta}h^{-\theta}}_{\text{marginal product of labour}}$$

$$\Rightarrow w = (1 - \theta)\frac{y}{h}$$

Worker's Problem

- ▶ The worker owns the capital
- ▶ Previously we had a budget constraint:

$$c = wh + \text{non labour income}$$

- ▶ Modify the budget constraint to include the government policy and return on capital
- ▶ The representative worker's utility maximisation problem:

$$\begin{cases} \max_{c,h} & \log c + \alpha \log(100 - h) \\ \text{s.t.} & (1 + \tau_c)c = (1 - \tau_h)wh + rk + \lambda \end{cases}$$

Worker's Problem: Solution

- ▶ Lagrangian with multiplier μ :

$$\mathcal{L} = \log c + \alpha \log(100 - h) + \mu [(1 - \tau_h)wh + rk + \lambda - (1 + \tau_c)c]$$

- ▶ FOCs:

$$\text{w.r.t. } c : \quad \frac{\partial \mathcal{L}}{\partial c} = 0 \quad \Rightarrow \quad \frac{1}{c} - \mu(1 + \tau_c) = 0$$

$$\text{w.r.t. } h : \quad \frac{\partial \mathcal{L}}{\partial h} = 0 \quad \Rightarrow \quad \alpha \frac{-1}{100 - h} + \mu(1 - \tau_h)w = 0$$

- ▶ Combining these, we have

$$\underbrace{\frac{1 - \tau_h}{1 + \tau_c}}_{\equiv (1 - \tau)} \frac{w}{c} = \alpha \frac{1}{100 - h}$$

Worker's Problem: Solution

- Therefore,

$$(1 - \tau) \frac{w}{c} = \alpha \frac{1}{100 - h}$$

where

$$\tau = \frac{\tau_c + \tau_h}{1 + \tau_c}$$

is the **effective tax on labor**

- The consumption tax τ_c creates a disincentive to work, as does the labour income tax τ_h
- High $\tau_c \Rightarrow$ Leisure is relatively less expensive $\Rightarrow$ Work less

Equilibrium Condition

- Substituting the condition for the wage w from the firm's problem,

$$(1 - \tau) \frac{w}{c} = \alpha \frac{1}{100 - h}$$

$$\Rightarrow (1 - \tau) \frac{(1 - \theta)y/h}{c} = \alpha \frac{1}{100 - h}$$

$$\Rightarrow (100 - h)(1 - \tau)(1 - \theta)y = \alpha ch$$

$$\Rightarrow h = \frac{100(1 - \theta)}{\frac{\alpha}{1 - \tau} \frac{c}{y} + (1 - \theta)}$$

Equilibrium Condition

$$h_{i,t} = \frac{100(1 - \theta)}{\frac{\alpha}{1 - \tau_{i,t}} \frac{c_{i,t}}{y_{i,t}} + (1 - \theta)}$$

- ▶ This is the core of the quantitative analysis in Prescott (2004)
- ▶ Use this to predict hours worked for each country i in period t
- ▶ We need:
 - ▶ Preference parameter α and production parameter θ
 - ▶ For each (i, t) , the consumption-output ratio $\frac{c_{i,t}}{y_{i,t}}$ and the effective tax rate $\tau_{i,t}$
- ▶ $\frac{c}{y}$ is an endogenous variable and should be solved for, but for simplicity we just take this from the data

Map the Model to the Data

Map the Model to the Data

- ▶ Want to predict hours worked $h_{i,t}$ for country i in period t
- ▶ Need the consumption-output ratio $\frac{c_{i,t}}{y_{i,t}}$ and the effective tax rate $\tau_{i,t}$
- ▶ What do they correspond to in the data?
- ▶ Since the model is very simple and abstracts from many things, this is a tricky task

System of National Accounts (SNA)

- ▶ Key source of data for macroeconomists
- ▶ The SNA follows the standard double-entry method of accounting
 - ▶ Lefthand side of the accounts: expenditure on final goods and services produced in the economy
 - ▶ Righthand side: the income earned by consumers, firms, and government
 - ▶ Entries on both sides must be balanced

Estimating Effective Tax Rate

- ▶ The government raises the money from taxes:
 - ▶ Direct taxes: taxes paid directly to the government, such as income and profit taxes. Not included in GDP
 - ▶ Indirect taxes: taxes paid indirectly to government, e.g. sales taxes and property taxes. Included in the prices of goods and services
- ▶ National income is equal to $GDP - IT$, where GDP and IT are the GDP and indirect taxes in the SNA

Estimating Effective Tax Rate: Consumption Tax τ_c

- ▶ IT consists of the indirect taxes on consumption IT_c and on investment IT_I , where $IT_c + IT_I = IT$
- ▶ Estimate the indirect taxes on consumption IT_c :

$$IT_c = \left(\frac{2}{3} + \frac{1}{3} \frac{C}{C + I} \right) IT$$

where

- ▶ C: consumption in the SNA
- ▶ I: investment in the SNA

Estimating Effective Tax Rate: Consumption Tax τ_c

$$IT_c = \left(\frac{2}{3} + \frac{1}{3} \frac{C}{C + I} \right) IT$$

- ▶ Consumption is about 2/3 of private expenditure
⇒ Assume that 2/3 of indirect taxes fall directly on consumption, representing the sales tax on consumption
- ▶ The remaining indirect taxes fall on business, which are partly passed on to consumers, e.g., gasoline or property taxes, which are reflected in higher prices for goods and services. Thus, the remaining 1/3 is divided between consumption and investment according to consumption's share of this spending

Estimating Effective Tax Rate: Consumption Tax τ_c

$$IT_c = \left(\frac{2}{3} + \frac{1}{3} \frac{C}{C + I} \right) IT$$

- ▶ Consumption in the SNA, C , includes the indirect taxes on consumption, so the actual consumption is $C - IT_c$
- ▶ The consumption tax rate is then calculated as

$$\underbrace{\tau_c}_{\text{tax rate}} \times \underbrace{(C - IT_c)}_{\text{consumption}} = \underbrace{IT_c}_{\text{indirect taxes on consumption}}$$

$$\Rightarrow \tau_c = \frac{IT_c}{C - IT_c}$$

Estimating Effective Tax Rate: Labour Income Tax τ_h

- ▶ The labour income tax consists of two taxes: a social security tax τ_{ss} and an income tax τ_{inc}
- ▶ The social security tax rate is proportional to labour income:

$$\underbrace{\tau_{ss}}_{\text{tax rate}} \times [\text{Labour Income}] = [\text{Social Security Taxes}]$$

$$\Rightarrow \tau_{ss} = \frac{\text{Social Security Taxes}}{\text{Labour Income}}$$

- ▶ Social Security Taxes are in the SNA
- ▶ What is labour income in the SNA?

Estimating Effective Tax Rate: Labour Income Tax τ_h

$$\tau_{ss} = \frac{\text{Social Security Taxes}}{\text{Labour Income}}$$

- ▶ Remember that the national income is GDP – IT. In the model, this corresponds to y
- ▶ With our production function, labour income is

$$\begin{aligned} wh &= \underbrace{(1 - \theta)zk^\theta h^{-\theta}}_{\text{marginal product of labour}} \times h \\ &= (1 - \theta)zk^\theta h^{1-\theta} \\ &= (1 - \theta)y \end{aligned}$$

- ▶ Thus labour income in the SNA is $(1 - \theta)(\text{GDP} - \text{IT})$
- ▶ Therefore

$$\tau_{ss} = \frac{\text{Social Security Taxes}}{(1 - \theta)(\text{GDP} - \text{IT})}$$

Estimating Effective Tax Rate: Labour Income Tax τ_h

- ▶ The *average* income tax rate is

$$\tau_{\text{inc}} = \frac{\text{Direct Taxes}}{\text{National Income}}$$

- ▶ In principle, income tax is levied on labour income, but Direct Taxes include taxes on interest income, and it is impossible to disentangle labor income taxes from taxes on interest income
- ▶ Thus use total income, not labour income, in the denominator

Estimating Effective Tax Rate: Labour Income Tax τ_h

- ▶ Now we have estimated τ_{ss} and τ_{inc}
- ▶ The estimate of the *marginal* tax rate on labour

$$\tau_h = \tau_{ss} + 1.6 \times \tau_{inc}$$

- ▶ Average of the marginal income tax rates are generally higher than average tax rates
- ▶ The factor 1.6 corresponds to this markup from average to marginal tax rates
- ▶ This is the most problematic number; it is also a strong assumption that this markup is the same across countries over time

Estimating Consumption-Output Ratio c/y

- ▶ Effective output in the model

$$y = \text{GDP} - \text{IT}$$

- ▶ Effective consumption in the model

$$c = C + (G - G_{\text{Mil}}) - \text{IT}_c$$

where $G - G_{\text{Mil}}$ is the nonmilitary government spending in the SNA

- ▶ Here assume that nonmilitary government spending is substitutable for private consumption in a one-to-one fashion

Estimating Parameters α and θ

- ▶ Recall that $(1 - \theta)(\text{GDP} - \text{IT})$ is the labour income in the SNA
- ▶ Thus, $1 - \theta$ is the labour income share, which gives $\theta = 0.32$
- ▶ In principle, θ can be estimated for each country, but assume it is common across countries
- ▶ What value should we use for α ?

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Calibration

Calibration

- ▶ Remember we needed
 - ▶ preference parameter α and production parameter θ
 - ▶ consumption-output ratio $\frac{c_{i,t}}{y_{i,t}}$ and effective tax rate $\tau_{i,t}$ for country i in period t
- ▶ Now we have c/y , τ and θ
- ▶ What value should we use for α ?

Calibration: Deep Parameter

- ▶ Deep parameters are the underlying structural parameters that govern the behaviour of economic agents
- ▶ These parameters are typically difficult to measure directly
- ▶ If the model is “true”, then the model should capture the main trade-off in reality, e.g., α will determine people's actual labour-leisure choices
- ▶ Thus, comparing the model's prediction with the data tells us what α should be
- ▶ Find α that minimises the model's prediction error

Calibration

- Find α that maximises the model's goodness of fit for the seven countries over both time periods:

$$\min_{\alpha} \sum_{i \in G7} \sum_{t=1,2} \left[h_{i,t}^{\text{data}} - h_{i,t}^{\text{model}}(\alpha) \right]^2$$

where $h_{i,t}^{\text{model}}(\alpha)$ is the model's predicted hours worked that is a function of α

- There are 7 countries $\times$ 2 points in time = 14 targeted moments

Calibration: Remarks

$$\min_{\alpha} \sum_{i \in G7} \sum_{t=1,2} \left[h_{i,t}^{\text{data}} - h_{i,t}^{\text{model}}(\alpha) \right]^2$$

- ▶ Calibration is sometimes ad hoc
- ▶ Prescott excluded two outlier observations, so he targeted 12 moments to obtain $\alpha = 1.54$
- ▶ What if we use the absolute value $|h^{\text{data}} - h^{\text{model}}(\alpha)|$ instead of the squared residuals?
- ▶ Some weights $\omega_{i,t}$ can be added

$$\min_{\alpha} \sum_{i \in G7} \sum_{t=1,2} \omega_{i,t} \left[h_{i,t}^{\text{data}} - h_{i,t}^{\text{model}}(\alpha) \right]^2$$

Calibration: Algorithm

- ▶ Step 1: load data
 - ▶ $\frac{c_{i,t}}{y_{i,t}}$ and $\tau_{i,t}$
 - ▶ Targeted moments $h_{i,t}$ (hours worked per person aged 15–64 per week)
- ▶ Step 2: set predetermined parameter(s), $\theta = 0.32$
- ▶ Step 3: calibrate α (using an optimisation routine)
- ▶ Step 4: plot the results
- ▶ Let's do a demonstration

Prediction and Welfare Analysis

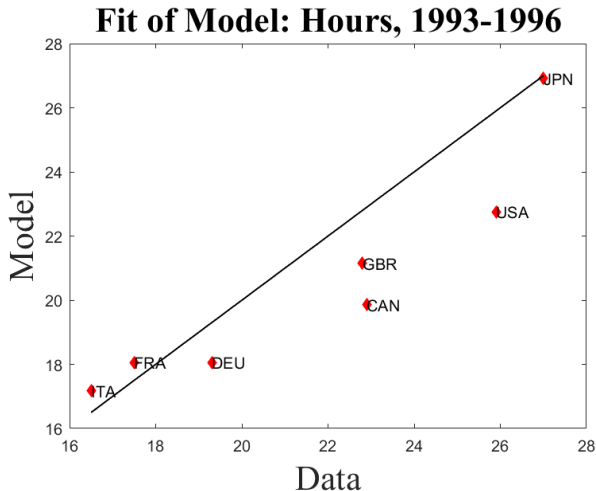
Actual and Predicted Labour Supplies

In Prescott (2004) where $\alpha = 1.54$ is used:

ACTUAL AND PREDICTED LABOR SUPPLY						
		<i>Labor Supply</i>		<i>Differences</i>	<i>Prediction Factors</i>	
		Actual	Predicted		τ	c/o
1993-96	Germany	19.3	19.5	0.2	.59	.74
	France	17.5	19.5	2.0	.59	.74
	Italy	16.5	18.8	2.3	.64	.69
	Canada	22.9	21.3	-1.6	.52	.77
	United Kingdom	22.8	22.8	0	.44	.83
	Japan	27.0	29.0	2.0	.37	.68
	United States	25.9	24.6	-1.3	.40	.81
1970-74	Germany	24.6	24.6	0	.52	.66
	France	24.4	25.4	1.0	.49	.66
	Italy	19.2	28.3	9.1	.41	.66
	Canada	22.2	25.6	3.4	.44	.72
	United Kingdom	25.9	24.0	-1.9	.45	.77
	Japan	29.8	35.8	6.0	.25	.60
	United States	23.5	26.4	2.9	.40	.74

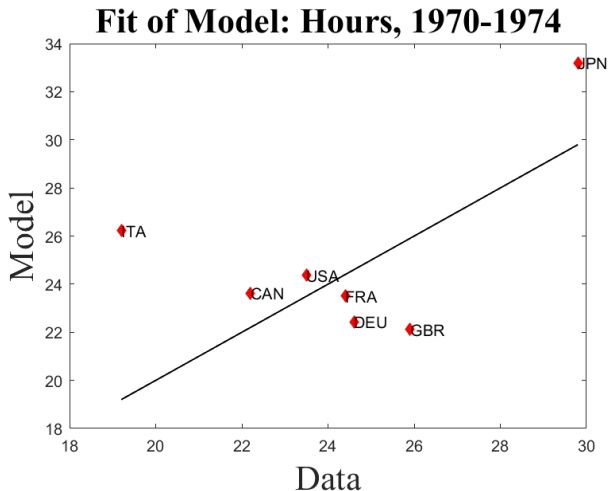
Actual and Predicted Labour Supplies

With the calibrated $\alpha = 1.71$,



Actual and Predicted Labour Supplies

With the calibrated $\alpha = 1.71$,



Model Performance

- ▶ The model predicts hours worked in the G7 well for the period 1993-96, and less well for the period 1970-74
- ▶ There may be other factors, which are not in the model, affecting the labour supply in the earlier period, e.g., labour market policies and demographics
- ▶ If the quantitative model is a good representation of the real world, we can do
 - ▶ Counterfactual analysis
 - ▶ Welfare analysis

Counterfactual Analysis

- ▶ Given the common parameters (α, θ) , countries are simply characterised by a combination of c/y and τ
- ▶ We assume the balanced growth path and thus that c/y remains constant (in the long-run) even if h changes
- ▶ Thus, we can simply plug a counterfactual τ into the equation for h to make a counterfactual prediction
- ▶ For example, for the UK in 1993-1996, $\tau_{UK,93-96} = 0.44$. If the UK were to raise its effective tax rate from this level to the French level of 59% ($\tau_{FRA,93-96} = 0.59$), labour supply would fall from 21.1 hours to 16.4 hours

Welfare Analysis (Ch 2.5)

- ▶ Want to calculate the welfare costs/gains of taxation
- ▶ Consider a switch from tax policy regime A to tax policy regime B
- ▶ Representative agent's welfare under policy regime A :

$$W_A \equiv U(c_A) + V(100 - h_A)$$

where c_A and h_A are consumption and labour supply under regime A

- ▶ Similarly, welfare under policy regime B :

$$W_B \equiv U(c_B) + V(100 - h_B)$$

- ▶ What are the welfare gains/losses of moving from regime A to B ?

Welfare Analysis

- ▶ Remember that utility is an ordinal (not cardinal) measure
- ▶ Thus, neither $\frac{W_B - W_A}{W_A}$ nor $\frac{W_B}{W_A}$ provides a meaningful measure of welfare gains/losses
- ▶ To see this, suppose we add a constant m to the utility function:

$$W_A \equiv U(c_A) + V(100 - h_A) + m$$

$$W_B \equiv U(c_B) + V(100 - h_B) + m$$

$\Rightarrow m$ does not affect the decision

- ▶ The order of welfare ($W_A \geq W_B$) does not change by this transformation. But, as $m \rightarrow \infty$,

$$\frac{W_B - W_A}{W_A} \rightarrow 0 \quad \text{and} \quad \frac{W_B}{W_A} \rightarrow 1$$

- ▶ Need a different concept

Welfare Analysis: Consumption Equivalent Variation

- ▶ **Consumption Equivalent Variation:** how much would a person be willing to pay, measured as a fraction of consumption in regime A , to move from A to B ?
- ▶ Given by ε that solves

$$U((1 + \varepsilon)c_A) + V(100 - h_A) = W_B$$

Welfare Analysis: Example

- ▶ Suppose the UK increases its effective tax rate from 44% to 59%
- ▶ Labour supply would fall from 21.1 hours to 16.4 hours
- ▶ This implies that leisure would increase from 78.9 hours to 83.6 hours
- ▶ Assuming the balanced growth path and a constant labor share, y decreases by the same factor as the labour supply; y becomes $16.4/21.1 \times y$
- ▶ Since c/y is constant, c would decrease by $16.4/21.1$

Welfare Analysis: Example

- ▶ Let the welfare of the UK in the counterfactual economy be

$$\begin{aligned} W &= U\left(\frac{16.4}{21.1}c_{UK}\right) + V(83.6) \\ &= \log\left(\frac{16.4}{21.1}c_{UK}\right) + \alpha \log(83.6) \end{aligned}$$

- ▶ Consumption Equivalent Variation: ε solves

$$\log((1 + \varepsilon)c_{UK}) + \alpha \log(100 - h_{UK}) = W$$

Welfare Analysis: Example

- Consumption Equivalent Variation: ε solves

$$\log((1 + \varepsilon)c_{UK}) + \alpha \log(100 - h_{UK}) = W$$

$$\Rightarrow \log(1 + \varepsilon) + \log(c_{UK}) + \alpha \log(78.9)$$

$$= \log\left(\frac{16.4}{21.1}\right) + \log(c_{UK}) + \alpha \log(83.6)$$

$$\Rightarrow \log(1 + \varepsilon) + \alpha \log(78.9) = \log\left(\frac{16.4}{21.1}\right) + \alpha \log(83.6)$$

$$\Rightarrow \log(1 + \varepsilon) = \log\left(\frac{16.4}{21.1}\right) + \alpha \log(83.6) - \alpha \log(78.9)$$

$$\Rightarrow \varepsilon = \exp\left[\log\left(\frac{16.4}{21.1}\right) + \alpha \log\left(\frac{83.6}{78.9}\right)\right] - 1$$

Welfare Analysis: Example

- ▶ This gives $\varepsilon = -0.142$
- ▶ Interpretation: If the UK had a higher effective tax rate (as high as in France), then people in the UK would enjoy more leisure time but consume less. Overall, the welfare of the UK people would decrease by 14.2 percent in terms of lifetime consumption equivalents

References

- ▶ Greenwood and Marto, Ch 2.3-2.5 and Ch 4
- ▶ Prescott “Why do Americans work so much more than Europeans?”, Quarterly Review, 2004